Does a 5/20-day moving-average rule beat buy-and-hold on Bitcoin?

Statistical inference with a t-test baseline and a moving block bootstrap

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One pre-specified rule, seven years of daily Bitcoin prices

2,557

daily returns, 2019–2025

5 / 20

day moving averages, fixed in advance

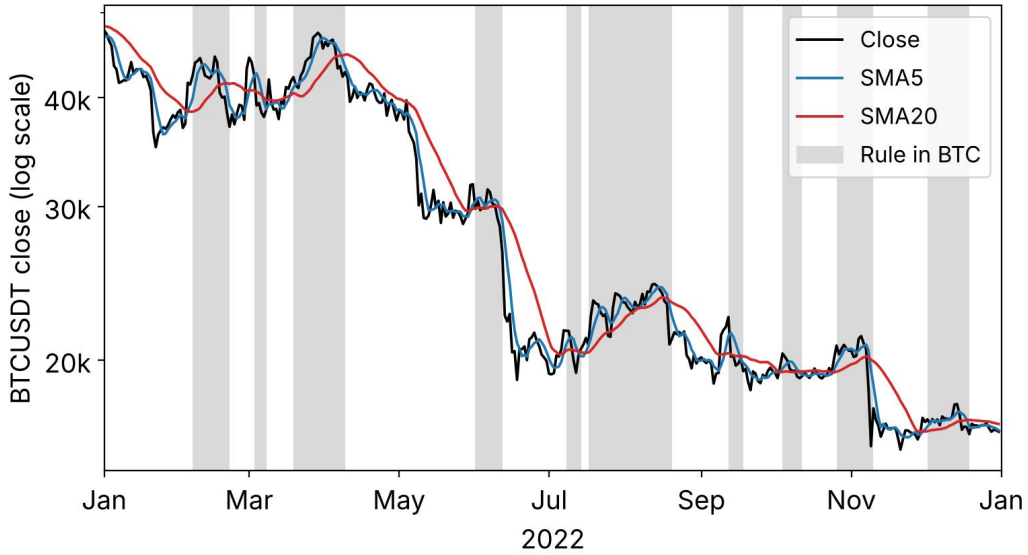
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parameters tuned on the data

Question and data

- Question: is the mean daily excess return d of the rule over buy-and-hold different from zero?
- Data: Binance public API, BTCUSDT spot, daily candles cut at 00:00 UTC; raw file and pull log in the repo

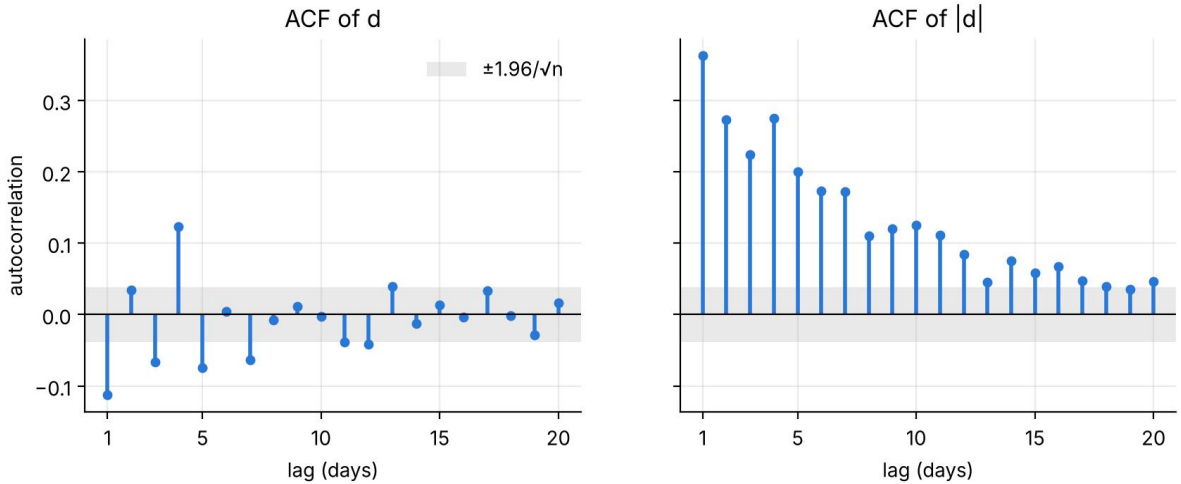
The rule decides each day using only yesterday's averages



Key points

- s = 1 if SMA5 > SMA20 on day t−1
- In BTC: earn r; out: hold USDT, earn 0
- d = −(1 − s) · r, so d = 0 on every day in BTC (55.4% of days)

The t-test assumes independent days, and the data say otherwise



- t-test baseline: t = −0.44, p = 0.66, so we cannot reject H0 (mean of d = 0)
- Independence fails: lag-1 autocorrelation of d is −0.11, outside the ±0.039 band, though the band assumes independence
- Negative autocorrelation makes the t-test SE too large: 0.000475 vs 0.000416

The block bootstrap keeps 14 consecutive days together

1 **Draw block starts**

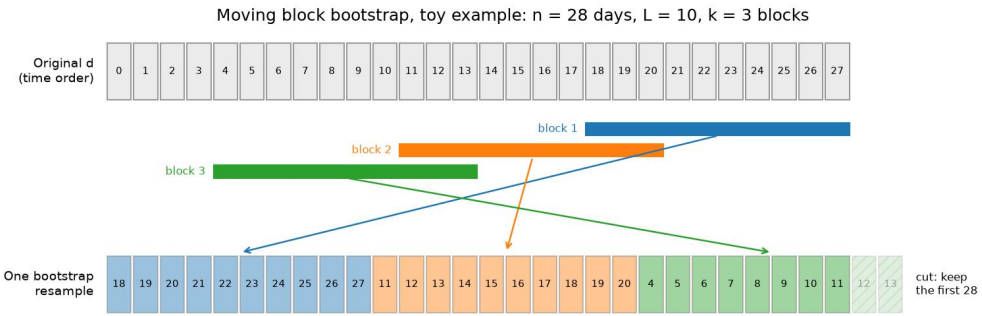
2,557 / 14, rounded up = 183 start days, drawn uniformly

2 **Take 14 days each**

14 consecutive days from each start, so neighbouring days stay together

3 **Stitch and cut to n**

join, cut 2,562 days back to 2,557, take the mean; repeat B = 9,999 times



Two intervals, same answer: the mean is not distinguishable from zero

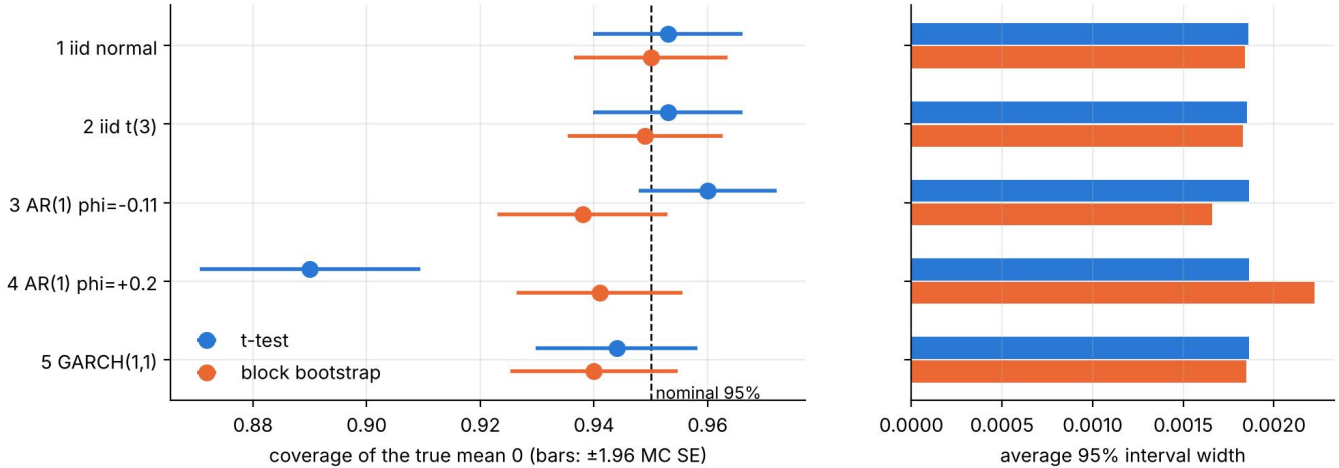
	t-test (i.i.d.)	Block bootstrap, l = 14
Mean of d	−0.00021	−0.00021
Standard error	0.000475	0.000416
95% CI	[−0.00114, 0.00072]	[−0.00101, 0.00062]
p-value	0.66	0.62

Source: outputs/tables/03_side_by_side.csv; bootstrap B = 9,999, seed 607

Key points

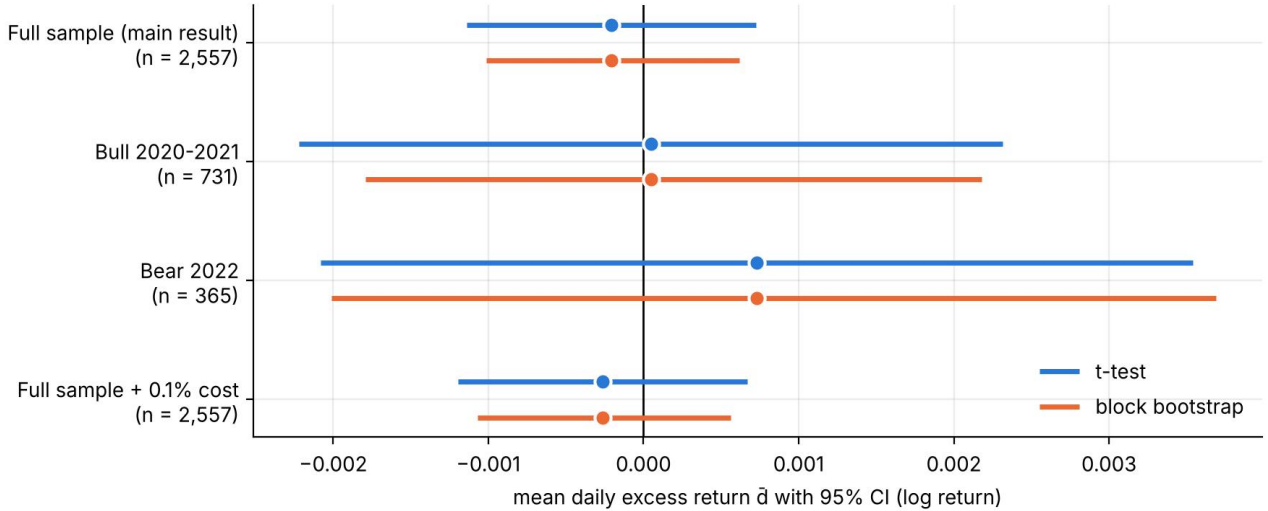
- Bootstrap SE is 12% smaller: d has lag-1 autocorrelation −0.11, and Ma's formula predicts SE 0.000407
- Both 95% CIs contain 0; for every block length from 1 to 40 the CI contains 0 (p = 0.58 to 0.66)

The t-interval under-covers only when days are positively correlated



- Five simulated series with true mean 0, 1,000 runs each
- At autocorrelation +0.2 the t-test covers only 89%, the bootstrap 94%
- Our d has the same sign of autocorrelation as DGP 3 (−0.11): t-test 96.0%, bootstrap 93.8%, both close to 95%

The answer holds in bull and bear years and after costs



- Bull 2020–2021: mean ≈ 0 (p = 0.97); bear 2022: about +27% a year (p = 0.61)
- With 0.1% cost per switch: about −9.6% a year, p = 0.58 (t-test), 0.53 (bootstrap)
- Every p > 0.5 and every CI contains 0; a robustness check, not a new result

What a decision-maker should take away

The rule's edge over buy-and-hold is statistically indistinguishable from zero

Result

Mean excess return −0.00021 per day (about −7.6% a year); 95% CI includes 0 under both methods

Method

Ignoring dependence changes the SE by about 12%, not the conclusion

Use

Treat a backtest gap of this size as noise until it beats this test

Limitations

One exchange

Binance BTCUSDT only

One rule

5/20-day crossover; other rules untested

Regime dependence

results differ across bull and bear years

Block length

l = 14 is a rule of thumb; checked from 1 to 40

Simplified costs

flat 0.1% per switch, no slippage

Sample window

2019–2025 only